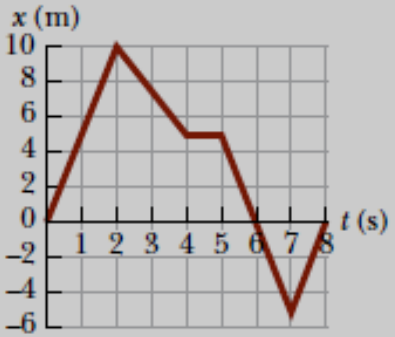
Problems and Solutions

Chapter 2

1. **W** The position versus time for a certain particle moving along the x axis is shown in Figure P2.1. Find the average velocity in the time intervals (a) 0 to 2 s, (b) 0 to 4 s, (c) 2 s to 4 s, (d) 4 s to 7 s, and (e) 0 to 8 s.



(a) $x = 0$ at $t = 0$ and $x = 10\text{ m}$ at $t = 2\text{ s}$:

$$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{10\text{ m} - 0}{2\text{ s} - 0} = \boxed{5.0\text{ m/s}}$$

(b) $x = 5.0\text{ m}$ at $t = 4\text{ s}$:

$$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{5.0\text{ m} - 0}{4\text{ s} - 0} = \boxed{1.2\text{ m/s}}$$

(c) $$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{5.0\text{ m} - 10\text{ m}}{4\text{ s} - 2\text{ s}} = \boxed{-2.5\text{ m/s}}$$

(d) $$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{-5.0\text{ m} - 5.0\text{ m}}{7\text{ s} - 4\text{ s}} = \boxed{-3.3\text{ m/s}}$$

(e) $$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{0.0\text{ m} - 0.0\text{ m}}{8\text{ s} - 0\text{ s}} = \boxed{0\text{ m/s}}$$

3. **M** A person walks first at a constant speed of 5.00 m/s along a straight line from point Ⓐ to point Ⓑ and then back along the line from Ⓑ to Ⓐ at a constant speed of 3.00 m/s. (a) What is her average speed over the entire trip? (b) What is her average velocity over the entire trip?

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}} = \frac{d_{AB} + d_{BA}}{t_{AB} + t_{BA}}$$

But $d_{AB} = d_{BA}$, $t_{AB} = d/v_{AB}$, and $t_{BA} = d/v_{BA}$

so $$\text{average speed} = \frac{d + d}{(d/v_{AB}) + (d/v_{BA})} = \frac{2(v_{AB})(v_{BA})}{v_{AB} + v_{BA}}$$

and

$$\text{average speed} = 2\left[\frac{(5.00\text{ m/s})(3.00\text{ m/s})}{5.00\text{ m/s} + 3.00\text{ m/s}}\right] = \boxed{3.75\text{ m/s}}$$

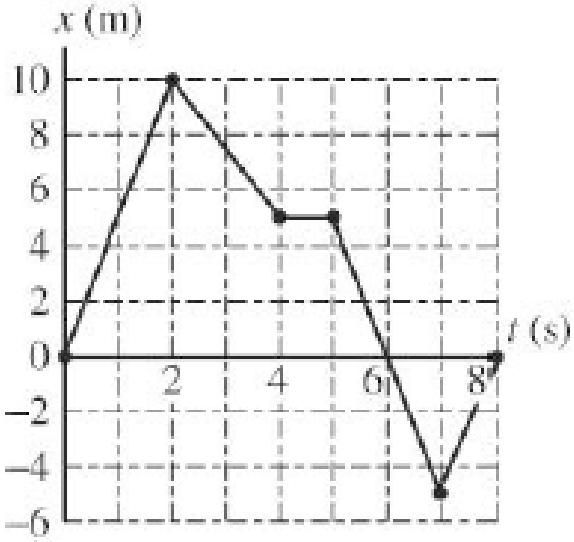
(b) The average velocity during any time interval equals total displacement divided by elapsed time.

$$v_{x,avg} = \frac{\Delta x}{\Delta t}$$

Since the walker returns to the starting point, $\Delta x = 0$ and $\boxed{v_{x,avg} = 0}$.

9. Find the instantaneous velocity of the particle **W** described in Figure P2.1 at the following times: (a) $t = 1.0$ s, (b) $t = 3.0$ s, (c) $t = 4.5$ s, and (d) $t = 7.5$ s.

The instantaneous velocity is found by evaluating the slope of the $x - t$ curve at the indicated time. To find the slope, we choose two points for each of the times below.



ANS. FIG. P2.9

(a) $$v = \frac{(5-0)\ \text{m}}{(1-0)\ \text{s}} = \boxed{5\ \text{m/s}}$$

(b) $$v = \frac{(5-10)\ \text{m}}{(4-2)\ \text{s}} = \boxed{-2.5\ \text{m/s}}$$

(c) $$v = \frac{(5-5)\ \text{m}}{(5\ \text{s}-4\ \text{s})} = \boxed{0}$$

(d) $$v = \frac{0-(-5\ \text{m})}{(8\ \text{s}-7\ \text{s})} = \boxed{+5\ \text{m/s}}$$

4. A particle moves according to the equation $x = 10t^2$, **W** where x is in meters and t is in seconds. (a) Find the average velocity for the time interval from 2.00 s to 3.00 s. (b) Find the average velocity for the time interval from 2.00 to 2.10 s.

We substitute for t in $x = 10t^2$, then use the definition of average velocity:

t (s)	2.00	2.10	3.00
x (m)	40.0	44.1	90.0

(a) $$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{90.0\ \text{m} - 40.0\ \text{m}}{1.00\ \text{s}} = \frac{50.0\ \text{m}}{1.00\ \text{s}} = \boxed{50.0\ \text{m/s}}$$

(b) $$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{44.1\ \text{m} - 40.0\ \text{m}}{0.100\ \text{s}} = \frac{4.10\ \text{m}}{0.100\ \text{s}} = \boxed{41.0\ \text{m/s}}$$

6. The position of a particle moving along the x axis varies in time according to the expression $x = 3t^2$, where x is in meters and t is in seconds. Evaluate its position (a) at $t = 3.00$ s and (b) at $3.00\text{ s} + \Delta t$. (c) Evaluate the limit of $\Delta x/\Delta t$ as Δt approaches zero to find the velocity at $t = 3.00$ s.

(a) At any time, t, the position is given by $x = (3.00\text{ m/s}^2)t^2$.

Thus, at $t_i = 3.00$ s: $x_i = (3.00\text{ m/s}^2)(3.00\text{ s})^2 = \boxed{27.0\text{ m}}$.

(b) At $t_f = 3.00\text{ s} + \Delta t$:: $x_f = (3.00\text{ m/s}^2)(3.00\text{ s} + \Delta t)^2$, or

$$x_f = \boxed{27.0\text{ m} + (18.0\text{ m/s})\Delta t + (3.00\text{ m/s}^2)(\Delta t)^2}$$

(c) The instantaneous velocity at $t = 3.00$ s is:

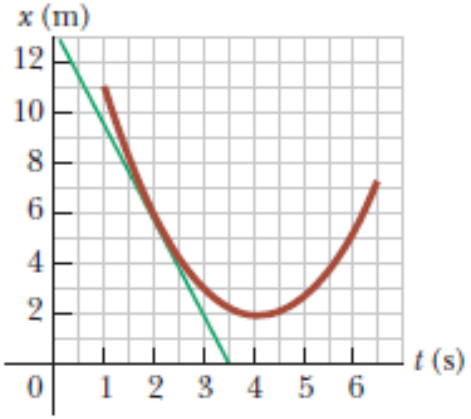
$$\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \lim_{\Delta t \to 0} \frac{x_f(t_i + \Delta t) - x_i(t_i)}{\Delta t}$$

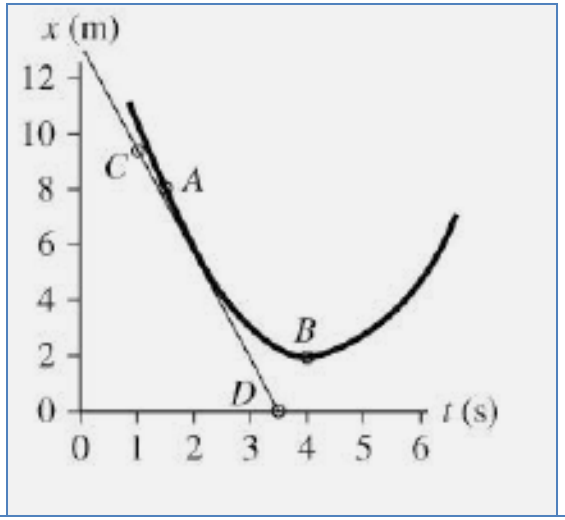
$$\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \lim_{\Delta t \to 0} \frac{(18.0\text{ m/s})\Delta t + (3.00\text{ m/s}^2)(\Delta t)^2}{\Delta t}$$

$$= \lim_{\Delta t \to 0} (18.0\text{ m/s}) + (3.00\text{ m/s}^2)(\Delta t) = \boxed{18.0\text{ m/s}}$$

$$or\ v = \frac{dx}{dt} = 6t = 6 * 3 = 18\ m/s$$

7. A position–time graph for a particle moving along the x axis is shown in Figure P2.7. (a) Find the average velocity in the time interval $t = 1.50$ s to $t = 4.00$ s. (b) Determine the instantaneous velocity at $t = 2.00$ s by measuring the slope of the tangent line shown in the graph. (c) At what value of t is the velocity zero?





(a) At $t_i = 1.5$ s, $x_i = 8.0$ m (Point A)

At $t_f = 4.0$ s, $x_f = 2.0$ m (Point B)

$$v_{avg} = \frac{x_f - x_i}{t_f - t_i} = \frac{(2.0 - 8.0)\ \text{m}}{(4.0 - 1.5)\ \text{s}}$$

$$= -\frac{6.0\ \text{m}}{2.5\ \text{s}} = \boxed{-2.4\ \text{m/s}}$$

(b) The slope of the tangent line can be found from points C and D. ($t_C = 1.0$ s, $x_C = 9.5$ m) and ($t_D = 3.5$ s, $x_D = 0$),

$$v \approx \boxed{-3.8\ \text{m/s}}$$

The negative sign shows that the **direction** of v_x is along the negative x direction.

(c) The velocity will be zero when the slope of the tangent line is zero. This occurs for the point on the graph where x has its minimum value. This is at $t \approx \boxed{4.0\ \text{s}}$.

15. A velocity–time graph for an object moving along the x axis is shown in Figure P2.15. (a) Plot a graph of the acceleration versus time. Determine the average acceleration of the object (b) in the time interval $t = 5.00$ s to $t = 15.0$ s and (c) in the time interval $t = 0$ to $t = 20.0$ s.

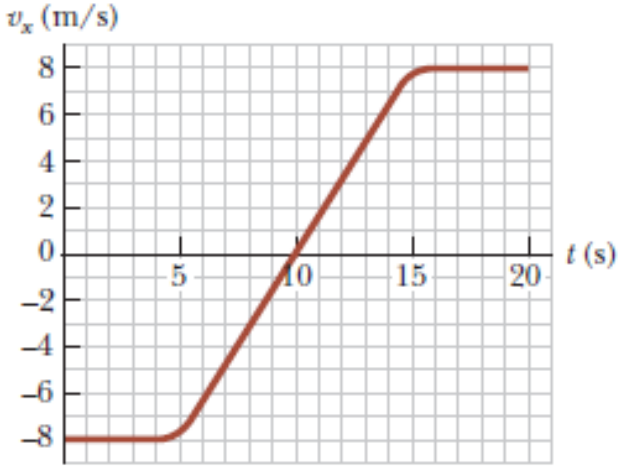


Figure P2.15

(a) Acceleration is the slope of the graph of v versus t.

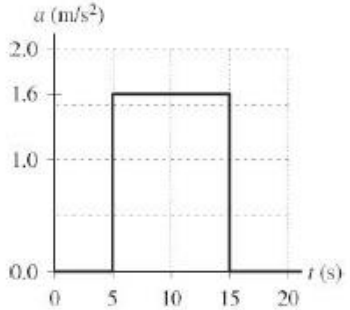
For $0 < t < 5.00$ s, $a = 0$.

For 15.0 s $< t <$ 20.0 s, $a = 0$.

For 5.0 s $< t <$ 15.0 s, $a = \dfrac{v_f - v_i}{t_f - t_i}$.

$$a = \frac{8.00\ \text{m/s} - (-8.00\ \text{m/s})}{15.0\ \text{s} - 5.00\ \text{s}} = \boxed{1.60\ \text{m/s}^2}$$

We can plot $a(t)$ as shown in ANS. FIG. P2.15 below.



ANS. FIG. P2.15

For (b) and (c) we use $a = \dfrac{v_f - v_i}{t_f - t_i}$.

(b) For 5.00 s $< t <$ 15.0 s, $t_i = 5.00$ s, $v_i = -8.00$ m/s, $t_f = 15.0$ s, and $v_f = 8.00$ m/s:

$$a = \frac{v_f - v_i}{t_f - t_i} = \frac{8.00\ \text{m/s} - (-8.00\ \text{m/s})}{15.0\ \text{s} - 5.00\ \text{s}} = \boxed{1.60\ \text{m/s}^2}$$

(c) We use $t_i = 0$, $v_i = -8.00$ m/s, $t_f = 20.0$ s, and $v_f = 8.00$ m/s:

$$a = \frac{v_f - v_i}{t_f - t_i} = \frac{8.00\ \text{m/s} - (-8.00\ \text{m/s})}{20.0\ \text{s} - 0} = \boxed{0.800\ \text{m/s}^2}$$

20. An object moves along the x axis according to the equation $x = 3.00t^2 - 2.00t + 3.00$, where x is in meters and t is in seconds. Determine (a) the average speed between $t = 2.00$ s and $t = 3.00$ s, (b) the instantaneous speed at $t = 2.00$ s and at $t = 3.00$ s, (c) the average acceleration between $t = 2.00$ s and $t = 3.00$ s, and (d) the instantaneous acceleration at $t = 2.00$ s and $t = 3.00$ s. (e) At what time is the object at rest?

(a) The average velocity is the change in position divided by the length of the time interval. We plug in to the given equation.

At $t = 2.00$ s, $x = [3.00(2.00)^2 - 2.00(2.00) + 3.00]$ m $= 11.0$ m.

At $t = 3.00$ s, $x = [3.00(3.00)^2 - 2.00(3.00) + 3.00]$ m $= 24.0$ m

so

$$v_{avg} = \frac{\Delta x}{\Delta t} = \frac{24.0 \text{ m} - 11.0 \text{ m}}{3.00 \text{ s} - 2.00 \text{ s}} = \boxed{13.0 \text{ m/s}}$$

(b) At all times the instantaneous velocity is

$$v = \frac{d}{dt}\left(3.00t^2 - 2.00t + 3.00\right) = \left(6.00t - 2.00\right)\ \text{m/s}$$

At $t = 2.00$ s, $v = [6.00(2.00) - 2.00]$ m/s = $\boxed{10.0\ \text{m/s}}$.

At $t = 3.00$ s, $v = [6.00(3.00) - 2.00]$ m/s = $\boxed{16.0\ \text{m/s}}$.

(c) $$a_{avg} = \frac{\Delta v}{\Delta t} = \frac{16.0\ \text{m/s} - 10.0\ \text{m/s}}{3.00\ \text{s} - 2.00\ \text{s}} = \boxed{6.00\ \text{m/s}^2}$$

(d) At all times $a = \frac{d}{dt}\left(6.00t - 2.00\right) = \boxed{6.00\ \text{m/s}^2}$. This includes both $t = 2.00$ s and $t = 3.00$ s.

(e) From (b), $v = (6.00t - 2.00) = 0 \rightarrow t = (2.00)/(6.00) = \boxed{0.333\ \text{s.}}$

21. M A particle moves along the x axis according to the equation $x = 2.00 + 3.00t - 1.00t^2$, where x is in meters and t is in seconds. At $t = 3.00$ s, find (a) the position of the particle, (b) its velocity, and (c) its acceleration.

To find position we simply evaluate the given expression. To find velocity we differentiate it. To find acceleration we take a second derivative.

With the position given by $x = 2.00 + 3.00t - t^2$, we can use the rules for differentiation to write expressions for the velocity and acceleration as functions of time:

$$v_x = \frac{dx}{dt} = \frac{d}{dt}\left(2 + 3t - t^2\right) = 3 - 2t \quad \text{and} \quad a_x = \frac{dv}{dt} = \frac{d}{dt}(3 - 2t) = -2$$

Now we can evaluate x, v, and a at $t = 3.00$ s.

(a) $x = (2.00 + 9.00 - 9.00)\ \text{m} = \boxed{2.00\ \text{m}}$

(b) $v = (3.00 - 6.00)\ \text{m/s} = \boxed{-3.00\ \text{m/s}}$

(c) $a = \boxed{-2.00\ \text{m/s}^2}$

25. An electron in a cathode-ray tube accelerates uniformly from 2.00×10^4 m/s to 6.00×10^6 m/s over 1.50 cm. (a) In what time interval does the electron travel this 1.50 cm? (b) What is its acceleration?

We have $v_i = 2.00 \times 10^4 \text{ m/s}$, $v_f = 6.00 \times 10^6 \text{ m/s}$, and $x_f - x_i = 1.50 \times 10^{-2} \text{ m}$.

(a) $x_f - x_i = \frac{1}{2}(v_i + v_f)t$:

$$t = \frac{2(x_f - x_i)}{v_i + v_f} = \frac{2(1.50 \times 10^{-2} \text{ m})}{2.00 \times 10^4 \text{ m/s} + 6.00 \times 10^6 \text{ m/s}}$$

$$= \boxed{4.98 \times 10^{-9} \text{ s}}$$

(b) $v_f^2 = v_i^2 + 2a_x(x_f - x_i)$:

$$a_x = \frac{v_f^2 - v_i^2}{2(x_f - x_i)} = \frac{(6.00 \times 10^6 \text{ m/s})^2 - (2.00 \times 10^4 \text{ m/s})^2}{2(1.50 \times 10^{-2} \text{ m})}$$

$$= \boxed{1.20 \times 10^{15} \text{ m/s}^2}$$

28. A truck covers 40.0 m in 8.50 s while smoothly slowing **W** down to a final speed of 2.80 m/s. (a) Find its original speed. (b) Find its acceleration.

(a) We use Equation 2.15:

$$x_f - x_i = \frac{1}{2}(v_i + v_f)t \text{ becomes } 40.0 \text{ m} = \frac{1}{2}(v_i + 2.80 \text{ m/s})(8.50 \text{ s}),$$

which yields $v_i = \boxed{6.61 \text{ m/s}}$.

(b) From Equation 2.13,

$$a = \frac{v_f - v_i}{t} = \frac{2.80 \text{ m/s} - 6.61 \text{ m/s}}{8.50 \text{ s}} = \boxed{-0.448 \text{ m/s}^2}$$

29. M An object moving with uniform acceleration has a velocity of 12.0 cm/s in the positive x direction when its x coordinate is 3.00 cm. If its x coordinate 2.00 s later is −5.00 cm, what is its acceleration?

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

and solve for a_x:

$$a_x = \frac{2\left[x_f - x_i - v_{xi}t\right]}{t^2}$$

We substitute:

$$a_x = \frac{2[-5.00\ \text{cm} - 3.00\ \text{cm} - (12.0\ \text{cm/s})(2.00\ \text{s})]}{(2.00\ \text{s})^2}$$

$$= \boxed{-16.0\ \text{cm/s}^2}$$

38. A particle moves along the x axis. Its position is given by the equation $x = 2 + 3t - 4t^2$, with x in meters and t in seconds. Determine (a) its position when it changes direction and (b) its velocity when it returns to the position it had at $t = 0$.

(a) Compare the position equation $x = 2.00 + 3.00t - 4.00t^2$ to the general form

$$x_f = x_i + v_i t + \frac{1}{2}at^2$$

to recognize that $x_i = 2.00$ m, $v_i = 3.00$ m/s, and $a = -8.00$ m/s^2. The velocity equation, $v_f = v_i + at$, is then

$$v_f = 3.00 \text{ m/s} - (8.00 \text{ m/s}^2)t$$

The particle changes direction when $v_f = 0$, which occurs at $t = \frac{3}{8}$ s. The position at this time is

$$x = 2.00 \text{ m} + (3.00 \text{ m/s})\left(\frac{3}{8} \text{ s}\right) - (4.00 \text{ m/s}^2)\left(\frac{3}{8} \text{ s}\right)^2$$

$$= \boxed{2.56 \text{ m}}$$

(b) From $x_f = x_i + v_i t + \frac{1}{2}at^2$, observe that when $x_f = x_i$, the time is given by $t = -\frac{2v_i}{a}$. Thus, when the particle returns to its initial position, the time is

$$t = \frac{-2(3.00\ \text{m/s})}{-8.00\ \text{m/s}^2} = \frac{3}{4}\ \text{s}$$

and the velocity is

$$v_f = 3.00\ \text{m/s} - \left(8.00\ \text{m/s}^2\right)\left(\frac{3}{4}\ \text{s}\right) = \boxed{-3.00\ \text{m/s}}$$

48. **W** A baseball is hit so that it travels straight upward after being struck by the bat. A fan observes that it takes 3.00 s for the ball to reach its maximum height. Find (a) the ball's initial velocity and (b) the height it reaches.

Since the ball's motion is entirely vertical, we can use the equations for free fall to find the initial velocity and maximum height from the elapsed time. After leaving the bat, the ball is in free fall for $t = 3.00$ s and has constant acceleration $a_y = -g = -9.80\ \text{m/s}^2$.

(a) The initial speed of the ball can be found from

$$v_f = v_i + at$$

$$0 = v_i - gt \rightarrow v_i = gt$$

$$v_i = (9.80\ \text{m/s}^2)(3.00\ \text{s}) = \boxed{29.4\ \text{m/s}}$$

(b) Find the vertical displacement Δy:

$$\Delta y = y_f - y_i = \frac{1}{2}(v_i + v_f)t$$

$$\Delta y = \frac{1}{2}(29.4\ \text{m/s} + 0)(3.00\ \text{s})$$

$$\Delta y = \boxed{44.1\ \text{m}}$$

50. The height of a helicopter above the ground is given by $h = 3.00t^3$, where h is in meters and t is in seconds. At $t = 2.00$ s, the helicopter releases a small mailbag. How long after its release does the mailbag reach the ground?

We are given the height of the helicopter: $y = h = 3.00t^3$.

At $t = 2.00$ s, $y = 3.00(2.00\text{ s})^3 = 24.0$ m and

$$v_y = \frac{dy}{dt} = 9.00t^2 = 36.0\text{ m/s}\uparrow$$

If the helicopter releases a small mailbag at this time, the mailbag starts its free fall with velocity 36.0 m/s upward. The equation of motion of the mailbag is

$$y_f = y_i + v_i t + \frac{1}{2}at^2$$

$$y_f = (24.0\text{ m}) + (36.0\text{ m/s})t - (4.90\text{ m/s}^2)t^2$$

Setting $y_f = 0$, dropping units, and rearranging the equation, we have

$$4.90t^2 - 36.0t - 24.0 = 0$$

We solve for t using the quadratic formula:

$$t = \frac{36.0 \pm \sqrt{(-36.0)^2 - 4(4.90)(-24.0)}}{2(4.90)}$$

Since only positive values of t count, we find $t = \boxed{7.96\text{ s}}$.

51. A ball is thrown directly downward with an initial **W** speed of 8.00 m/s from a height of 30.0 m. After what time interval does it strike the ground?

The equation for the height of the ball as a function of time is

$$y_f = y_i + v_i t - \frac{1}{2} g t^2$$

$$0 = 30\ \text{m} + (-8.00\ \text{m/s})t - (4.90\ \text{m/s}^2)t^2$$

Solving for t,

$$t = \frac{+8.00 \pm \sqrt{(-8.00)^2 - 4(-4.90)(30)}}{2(-4.90)} = \frac{+8.00 \pm \sqrt{64 + 588}}{-9.80}$$

$$t = \boxed{1.79\ \text{s}}$$

52. A ball is thrown upward from the ground with an initial speed of 25 m/s; at the same instant, another ball is dropped from a building 15 m high. After how long will the balls be at the same height above the ground?

M

The falling ball moves a distance of (15 m – h) before they meet, where h is the height above the ground where they meet. We apply

$$y_f = y_i + v_i t - \frac{1}{2} g t^2$$

to the falling ball to obtain

$$-(15.0\ \text{m} - h) = -\frac{1}{2} g t^2$$

or $$h = 15.0\ \text{m} - \frac{1}{2} g t^2 \qquad \text{[1]}$$

Applying $y_f = y_i + v_i t - \frac{1}{2} g t^2$ to the rising ball gives

$$h = (25\ \text{m/s}) t - \frac{1}{2} g t^2 \qquad \text{[2]}$$

Combining equations [1] and [2] gives

$$(25\ \text{m/s}) t - \frac{1}{2} g t^2 = 15.0\ \text{m} - \frac{1}{2} g t^2$$

or $$t = \frac{15\ \text{m}}{25\ \text{m/s}} = \boxed{0.60\ \text{s}}$$